ECOSOAR BUILD, SET UP & FLIGHT OPERATIONS

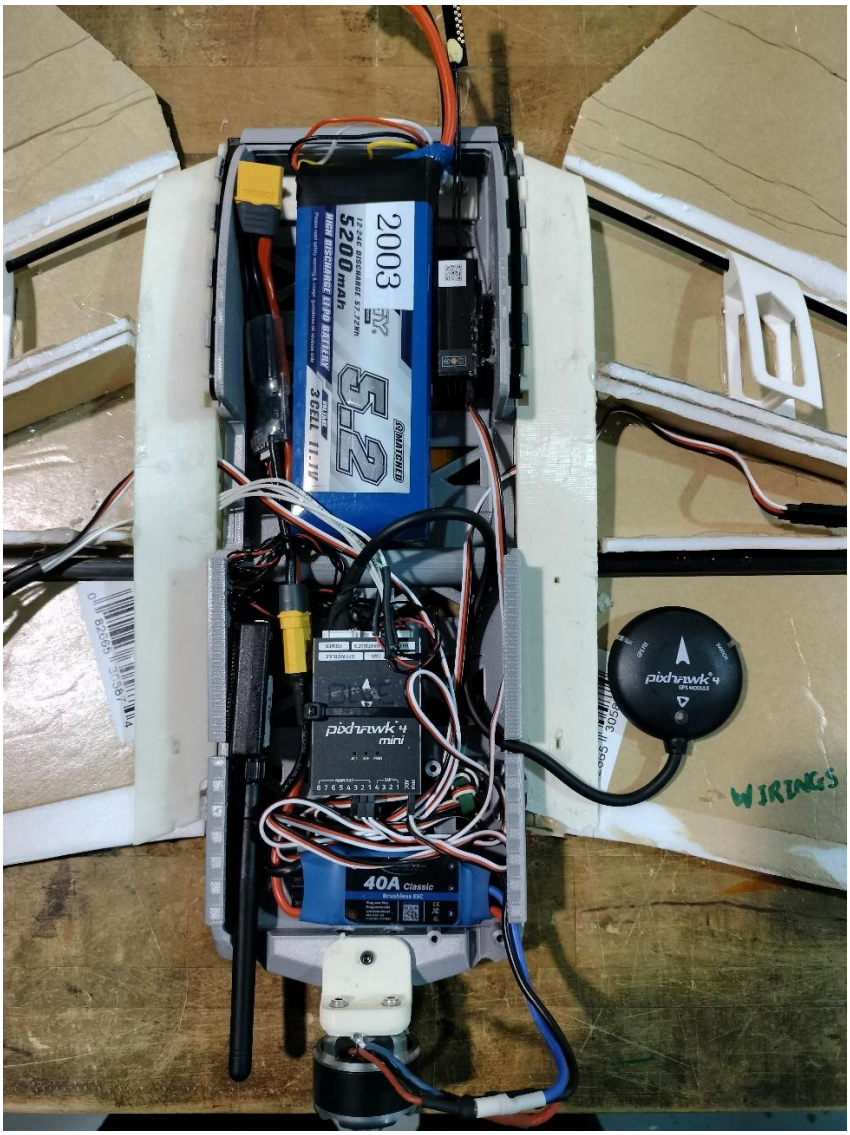
BUILD PART

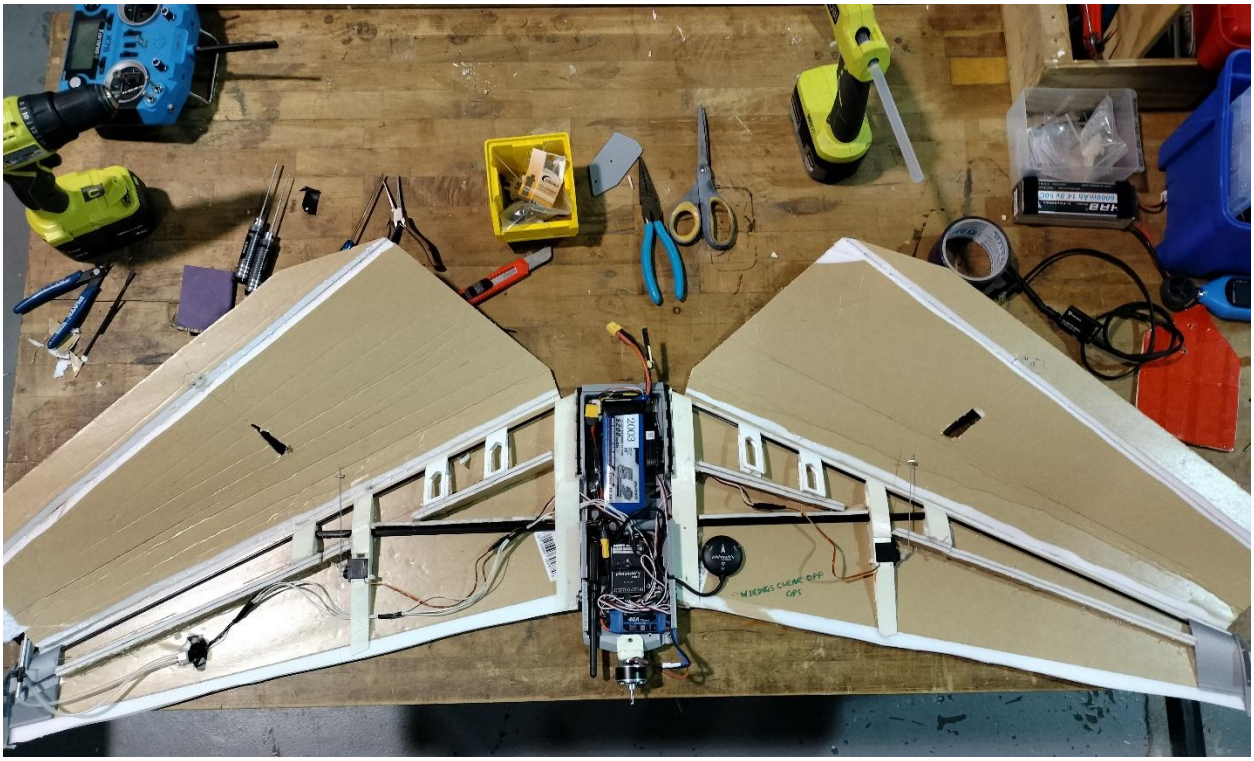
1. Rear fuselage sections side panels height must be increased so that the cover on top can be slid through it. The electronics and wiring hinder the top cover passage.

SET UP

ELECTRONICS PLACEMENT

1. The electronics must be placed as shown in picture below. The placement of electronics in such manner helps in balancing the center of gravity (especially in fuselage).



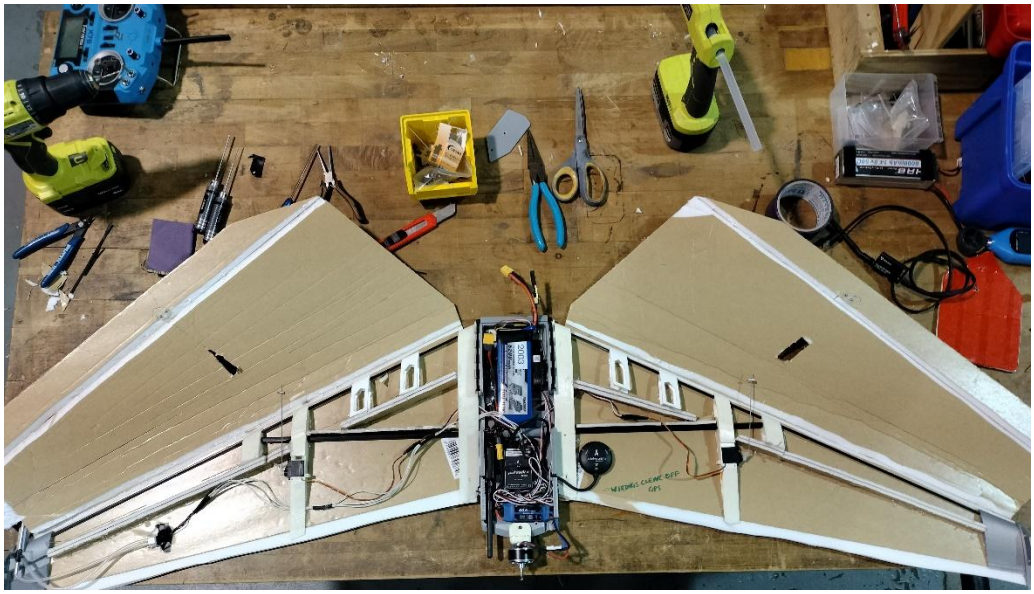


2. The servo should be calibrated before operation. Watch the video attached in the link below to set the neutral point of servo as 1500 µs.
   NOTE: - Skip the first 6 min and watch the rest.
   Calibrating the servo to neutral point.

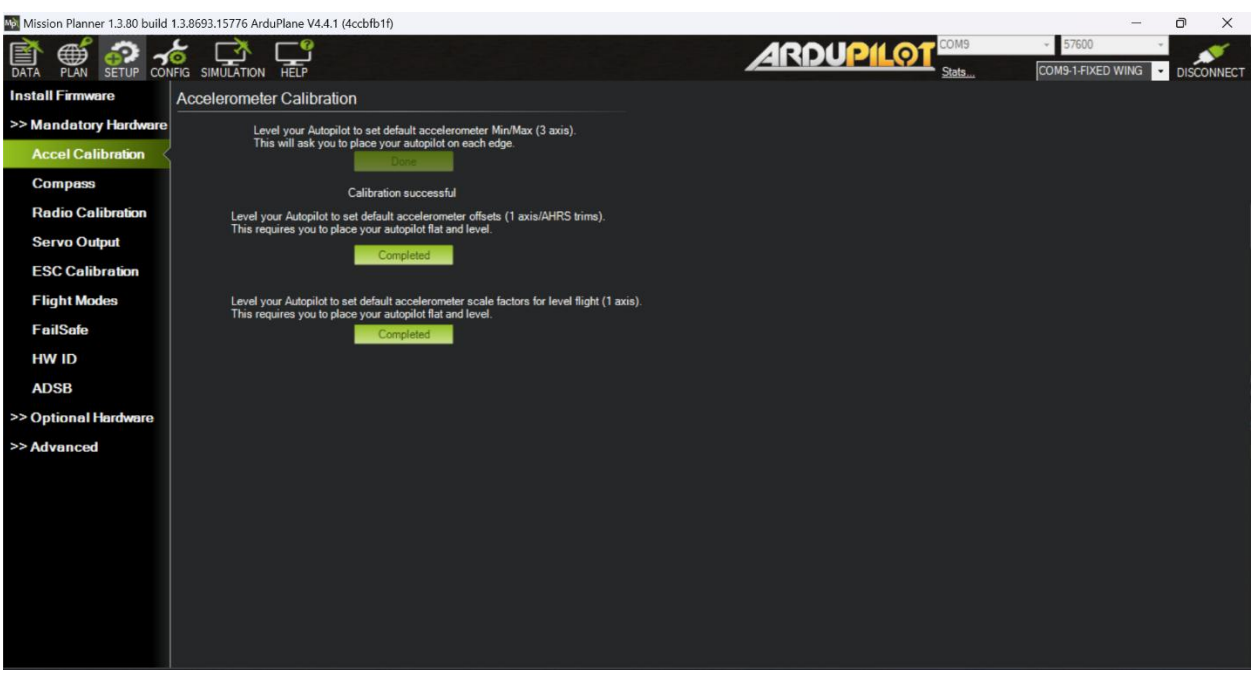
3. Calibrate the airspeed sensor as per the documentation provided in the link below.
   NOTE: - Only perform what is mentioned under the heading ‘Preflight’ when you open the link given below.
   Calibrating airspeed sensor

4. The accelerometer must be calibrated by removing the winglets attached on the wings. This is done since the flight controller should be placed flat on level surface while calibrating the accelerometer.

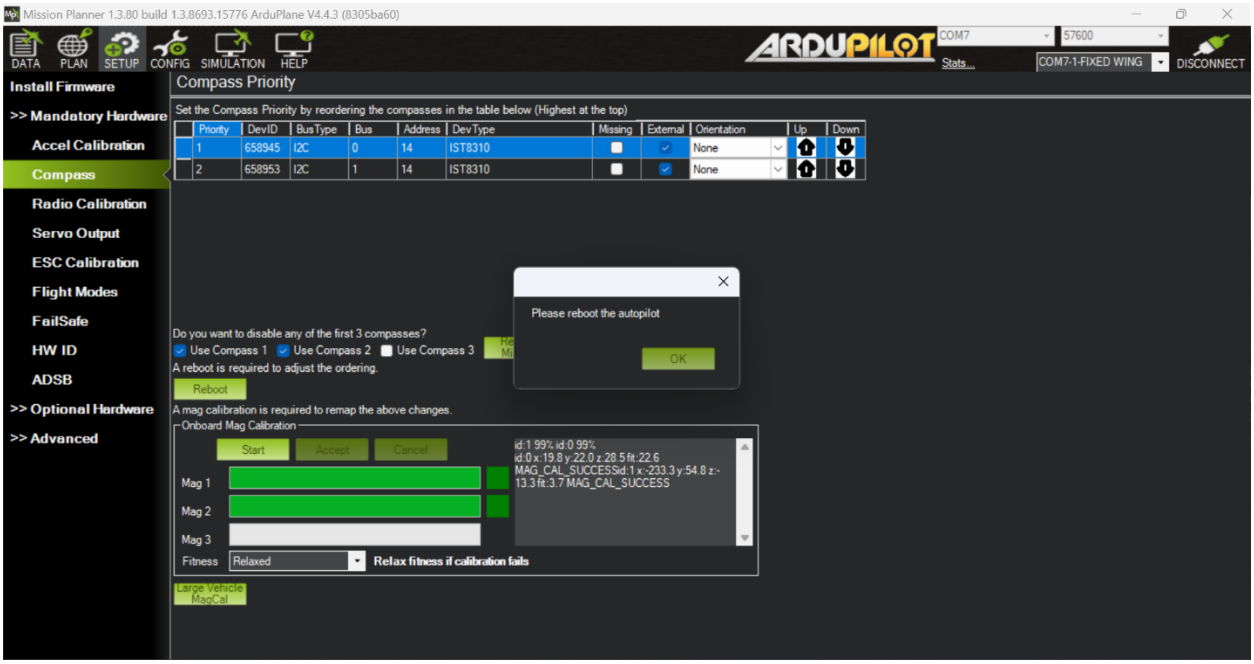
   Picture shown below does not have any winglet on the sides. However, the wings must be closed before performing accelerometer calibration.



   The Accel calibration option must show ‘Done’, ‘Completed’, ‘Completed’ in their respective section as seen in the picture below.

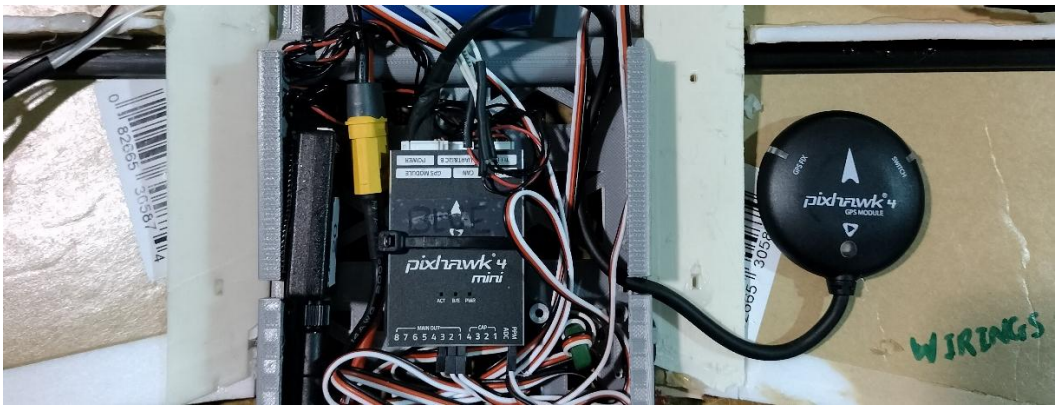


5. Magnetometer calibration must be performed in an open area away from anything that induces emf.



Both internal and external compass calibration must be fully finished to achieve perfect heading while doing the flight test.

The external GPS + Compass module must be aligned with the direction of the flight controller as seen in the picture below.



If it is not aligned, then proper rotation must be given in the parameter COMPASS_ORIENT

6. Parameters to be updated in Mission Planner are provided in the excel sheet attached below.
   [Ecosoar updated parameter.xlsx]

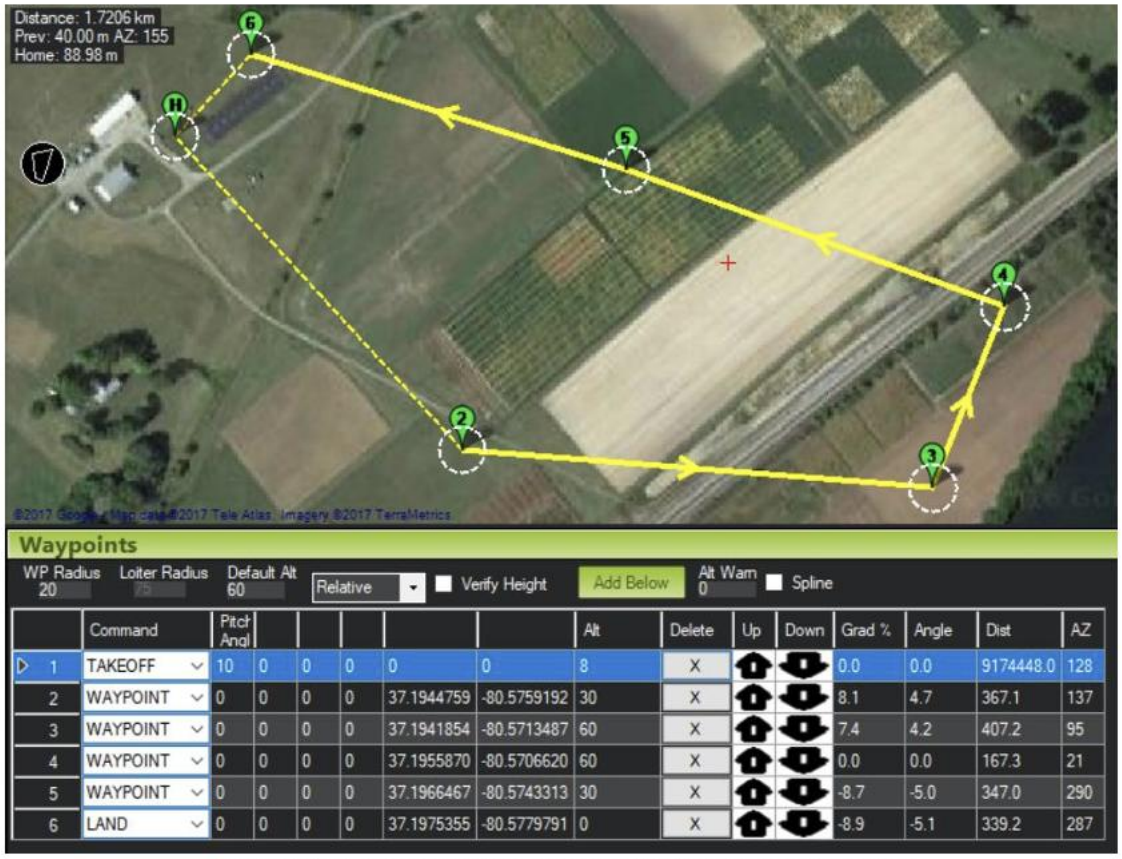
FLIGHT OPERATIONS

1. Flight Mission can be set as shown in the image below.

The 'Do change speed' row can be experimented with; its addition or removal did not make lot of difference.

The white circles around the yellow straight line indicate WP_RADIUS. It was set as 90 on the first and 2nd day of flight. However, it must be reduced to 10 or 20 for straight and short missions.

2. Long missions with multiple waypoints in a polygon shape can be as shown in image below. While carrying out the following mission make sure that the WP_RADIUS is 20m or a little more.



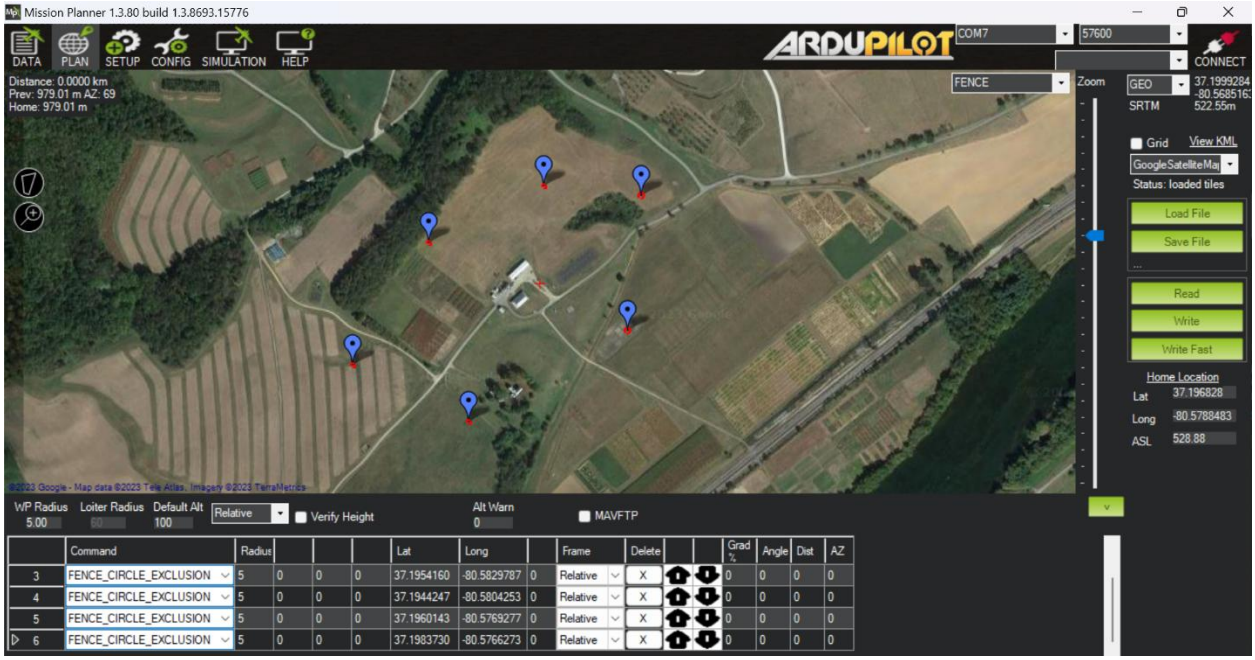
	Command	Pitch Angl						Alt	Delete	Up	Down	Grad %	Angle	Dist	AZ
1	TAKEOFF	10	0	0	0	0	0	8	X			0.0	0.0	9174448.0	128
2	WAYPOINT	0	0	0	0	37.1944759	-80.5759192	30	X			8.1	4.7	367.1	137
3	WAYPOINT	0	0	0	0	37.1941854	-80.5713487	60	X			7.4	4.2	407.2	95
4	WAYPOINT	0	0	0	0	37.1955870	-80.5706620	60	X			0.0	0.0	167.3	21
5	WAYPOINT	0	0	0	0	37.1966467	-80.5743313	30	X			-8.7	-5.0	347.0	290
6	LAND	0	0	0	0	37.1975355	-80.5779791	0	X			-8.9	-5.1	339.2	287

Figure 36. Sample mission showing all required waypoints

3. Make sure that all arming checks are ON in the parameters. Resolve if there is any error shown on the Mission planner Heads Up Display (HUD).

4. Sometimes the error could be 'Fence set but none selected'. For resolving this issue, go to the 'Plan' section and on the right top corner select 'Fence' by toggling the options and set a polygon fence around your mission. Considerable Margin must be provided between 'Mission waypoints' and 'Fence polygon'.

   Check the image shown below for reference.



5. Consider arming the drone once it is set on the launcher and then writing the mission plan as per its heading.